Irtifa Haider

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RESEARCH INTEREST

Natural Language Processing (NLP); Large Language Models (LLMs); LLM Reliability: Machine Learning; Information Retrieval; Data Mining

EDUCATION

Master of Science (M.Sc.) in Computer Science and Engineering **Sept 2025 – Present**
Jahangirnagar University, Savar, Dhaka, Bangladesh

Bachelor of Science (B.Sc.) in Computer Science and Engineering **Mar 2020 – Sept 2025**
Jahangirnagar University, Savar, Dhaka, Bangladesh

- **CGPA:** 3.43/4.00
- Thesis: *The Star Paradigm: Redefining Social Network Clustering*.

RESEARCH EXPERIENCE PUBLICATION

- **Haider, I.**, Alam, S., Hossan, M. T., Anwar, M. M., & Aurpa, T. T. A Comprehensive Text Optimization Approach to Bangla Summarization. *Proceedings of the Second Bangla Language Processing Workshop (BLP), co-located with the 14th International Joint Conference on Natural Language Processing (IJCNLP–AACL)*, 2025. [Paper]

- **The Star Paradigm: Redefining Social Network Clustering.** *Manuscript in preparation.*
  - Proposed and evaluated a graph-based star-clustering approach for social network analysis.
  - Compared clustering variants with baseline methods on classic networks and LFR benchmark graphs using modularity, coverage, NMI, runtime, and community count.
  - Investigated incremental node arrivals, sensitivity to parameter τ, scalability, node-arrival order, and robustness to noisy edges.

- **HSECRS: Hybrid Smart Energy Control and Redistribution System.** *Manuscript in preparation.*
  - Developing an AI-enabled smart energy management and redistribution system for efficient power utilization, with an IoT prototype that optimizes power distribution across solar, grid, and battery sources while prioritizing critical loads.

PROFESSIONAL EXPERIENCE

QA Functional Tester (Internship) | A1QA **April 2026 – June 2026**

- Worked with various testing types including Smoke Testing, Acceptance Testing, and Manual Testing.
- Designed and executed test cases, performed test planning, identified bugs, and reported defects using Jira with proper documentation.

Coding Instructor | CoderPals **Sept 2024 – April 2026**

- Taught Scratch and Python through engaging, game-based, and project-based lessons for children.
- Simplified programming concepts to foster computational thinking, creativity, and problem-solving skills.

ACADEMIC PROJECTS

Plant Classification | Python, scikit-learn, scikit-image, NumPy, pandas, matplotlib, joblib

- Built a machine-learning pipeline to classify 18 plant types using HOG features and SGD/SVM classifiers.
- Achieved 80% classification accuracy through feature extraction and model tuning.

Hybrid Smart Energy System (IoT Project) | C++, Arduino IDE, ESP32, IoT Sensors, Blynk, Solar Panel

- Designed a cost-effective IoT system to optimize power distribution across solar, grid, and battery sources.

- Prioritized critical loads to improve energy efficiency and support sustainable power management.

LivSmart: Smart Living Community App | Java, XML, Firebase, Dokka, JUnit, Mockito, MVVM

- Developed an Android application to streamline residential services, including service requests, payments, events, and alerts.
- Integrated a Firebase backend for dashboards, parking, and amenities booking.

OpenDoor: Property Rental & Purchase App | Java, XML, Firebase, CircleCI, JUnit, Mockito, MVVM

- Built an Android app to simplify property rental and purchase through aggregated listings and filtering.
- Added real-time alerts and a mortgage calculator to improve the property-search experience.

Learn and Teach Programming | C++

- Developed a C++ learning platform with tutorials, code examples, and lessons to support programming practice.

Criss Cross Game & Calculator | C

- Built a two-player Tic-Tac-Toe game with an integrated calculator for basic arithmetic operations.

IT Training Hub | HTML, CSS, JavaScript, PHP, MySQL, MVC

- Developed a web platform for managing IT training courses, trainers, and participants.
- Applied the Model–View–Controller (MVC) architecture to organize the application and support maintainability.

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JavaScript, PHP

AI Frameworks & Models: TensorFlow, PyTorch, scikit-learn, NumPy, pandas, matplotlib, Word2Vec, BERT, BanglaBERT, mBERT, XLM-RoBERTa (XLM-R), mT5, T5, GPT-4, Gemini, Llama, Qwen2.5

Web Development: HTML, CSS, React, Bootstrap, Django, REST API, Amazon Web Services (AWS)

Development Tools: Git, GitHub, Android Studio, VS Code, NetBeans, Eclipse, Anaconda, Visual Paradigm, CI/CD, JUnit, Mockito, MATLAB, Slack, Trello, MySQL, MongoDB, Firebase, Blynk

Productivity & Collaboration Tools: Microsoft Office Suite (Word, Excel, PowerPoint), Toggl, Overleaf, JIRA

Certifications & Professional Development

- **Certificate of Attendance** – IJCNLP–AACL 2025 **Jan 2026**
- **Cython Codecraft: Data Structure & Algorithm Bootcamp** – bongoDev – 3-Day Bootcamp **Feb 2024**
- **Learn & Hack Web Development Bootcamp** –bongoDev – 2-Week Bootcamp **Jan 2024**
- **Legacy JavaScript Algorithms and Data Structures** – freeCodeCamp **Aug 2023**
- **Responsive Web Design** – freeCodeCamp **Jul 2023**

HONORS & AWARDS

- **CP:** Solved 400+, Codeforces (*Irtifa_CSE-29*), AtCoder (*Irtifa_CSE29*), and LeetCode (*haiderIr*).
- **International Conference:** Presented paper at the *14th International Joint Conference on Natural Language Processing (IJCNLP) & 4th Asia-Pacific Chapter of the Association for Computational Linguistics*. **2025**
- **International Collegiate Programming Contest (ICPC):** Competed in the preliminary round. ***2021***
- **National Collegiate Programming Contest (NCPC):** Participated in the preliminary round. ***2021***
- **Merit Scholarships:** Received Government Board Merit Scholarships in both *JSC (2014)* and *SSC (2017)*.

VOLUNTEER EXPERIENCE

Volunteer | Deep Ecology and Snake Conservation Foundation **2023 – Present**

- Participate in environmental conservation initiatives, awareness campaigns, and community outreach programs.

Event Support Staff | National Collegiate Programming Contest (NCPC), Jahangirnagar University **2024**

- Assisted in coordinating the on-site logistics and smooth execution of the NCPC 2024.
- Supported venue management, and communication with committees to ensure successful event operations.